REPORT

On

Micro Project – ADVANCED PYTHON PROGRAMMING

TOPIC NAME : E-COMMERCE SALES AND PRODUCT ANALYTICS DASHBOARD

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In

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INDEX

Sl. No.	Particulars	Page No.
1	Rationale	
2	Competency of the Course	
3	Course Outcomes Addressed	
4	Information Collection	
5	Theory and Design of the Micro Project	
6	Methodology Followed	
7	Outputs of the Micro–project	
8	Benefits of this Micro–project	
9	Area of Future Improvement	
10	References	

1.Rationale

E-commerce businesses produce a large amount of transactional data every day. Information such as order numbers, product names, categories, quantities, prices and dates must be organized before it can be analysed effectively. A business that only stores raw records may find it difficult to identify its best-selling products, high-performing categories and overall sales trends. The E-Commerce Sales and Product Analytics Dashboard is proposed to solve this problem by converting structured sales records into useful analytical summaries. The project applies Data Structures concepts to a realistic business situation. Lists or arrays can hold records, dictionaries or hash tables can support fast key-based lookup and aggregation, and sorting techniques can rank products according to revenue or quantity sold. The dashboard combines these processed results into understandable metrics and tables. Thus, the project provides practical experience in handling data while demonstrating how organized data can support business analysis and decision-making.

2.Competency of the Course

The project develops competency in selecting and applying appropriate data structures and algorithms for a practical data-processing problem. Students learn to represent e-commerce records systematically, perform operations such as insertion, traversal, searching and sorting, and generate aggregated information from multiple records. The project also encourages analysis of how the choice of a data structure affects the efficiency and clarity of the solution.

3.Course Outcomes Addressed in this Micro Project

CO1: Apply fundamental data structures for representing, organizing and managing data.
CO2: Implement searching, sorting and basic data-processing operations using suitable data structures.
CO3: Analyse the suitability and efficiency of data structures and algorithms for a real-world application.

4.0 Information Collection

Information required for the project was identified from common e-commerce sales and product analysis requirements. A representative sales record may contain an order ID, product ID, product name, category, quantity, unit price and order date. These fields allow the system to calculate revenue, compare products and study category performance. The project focuses on four major types of information: transaction-level details, product-level details, category-level summaries and overall business indicators. During processing, revenue for an individual record is calculated by multiplying quantity by unit price. Product and category totals are then obtained by accumulating values using suitable keys. Searching can be used to find a particular product or order, while sorting can arrange products by sales value or quantity. Unique identifiers can be maintained using a set. The collected information is finally transformed into dashboard metrics, rankings and tables. Data validation is also considered because missing prices, invalid quantities, duplicate identifiers or inconsistent categories may produce incorrect analytical results.

1.Theory and Design of the Micro Project
2.System Design

The proposed system follows a sequence of stages: Input Sales Data ® Data Validation ® Data Structure Storage ® Processing ® Analytics ® Dashboard Output. First, sales records are collected and checked for basic consistency. Valid records are stored using appropriate data structures. During processing, the system traverses the records and calculates revenue and summary values. Product and category totals are maintained using key-based aggregation. Finally, the processed results are displayed through dashboard indicators, tables and rankings. This design separates data storage, processing and presentation, making the system easier to understand and modify.

1.Data Structures Used

Data Structure	Role in the Project
List / Array	Stores sales records and allows sequential traversal and processing.
Dictionary / Hash Table	Stores product/category keys with totals and supports fast lookup and aggreg
Set	Maintains unique product IDs, order IDs or category values.
Stack / Queue	Can represent recent user actions or pending data-processing tasks when req

5.Core Processing and Algorithms

- **Revenue calculation: For every transaction, multiply quantity by unit price and add the result to the required totals.**
- **Aggregation: Use product and category keys to maintain total revenue and units sold.**
- **Searching: Search by product ID, product name or order ID to retrieve a required record.**
- **Sorting: Sort product summaries by revenue or quantity to obtain a top-selling list.**
- **Dashboard calculation: Derive total sales, total orders, units sold and average order value from processed records.**

6.Methodology Followed

- **Identify the e-commerce sales problem and define the required analytical measures.**
- **Design the fields of a sales record and select appropriate data structures.**
- **Prepare a representative dataset and check for missing, duplicate or invalid values.**
- **Store the records and implement traversal, searching and aggregation operations.**
- **Apply sorting to generate product rankings and category comparisons.**
- **Calculate dashboard indicators and prepare structured output tables.**
- **Review the output for consistency and document findings, limitations and possible improvements.**

7. Outputs of the Micro–project
7.1 Dashboard Summary

TOTAL SALES	TOTAL ORDERS	UNITS SOLD	AVG. ORDER VALUE
■ 2,45,800	1,240	3,685	■ 198

7.2 Top Product Analysis

Rank	Product	Category	Units	Revenue
1	Wireless Headphones	Electronics	420	■ 84,000
2	Smart Watch	Electronics	315	■ 63,000
3	Backpack	Fashion	280	■ 42,000
4	Running Shoes	Sports	190	■ 38,000
5	Water Bottle	Home & Fitness	160	■ 18,800

7.3 Findings and Interpretation

- **The structured representation makes it easier to retrieve and process individual sales records.**
- **Key-based aggregation is suitable for calculating product-wise and category-wise totals.**
- **Sorting helps management identify products that contribute strongly to sales.**
- **Summary indicators reduce the effort required to interpret a large number of transaction records.**
- **Product and category comparisons can reveal where sales are concentrated.**
- **Data quality is essential because incorrect quantity or price values directly affect revenue and rankings.**

7.4 Conclusion

The micro-project successfully demonstrates a practical connection between Data Structures and e-commerce analytics. The proposed design organizes sales information, processes it through searching, sorting and aggregation, and presents the results in dashboard form. The project shows that selecting suitable structures can make data processing more systematic and that processed data can be converted into useful business information.

7.5 Limitations

- **The project uses a representative dataset and does not cover every complexity of a production e-commerce platform.**
- **Real-time transaction streaming, payment processing and customer-level recommendation systems are outside the current scope.**
- **Analytical accuracy depends on the completeness, correctness and consistency of the input dataset.**

8. Benefits of this Micro–project

This micro-project gives students practical experience in applying Data Structures to a real-world business problem. It strengthens understanding of lists, dictionaries, sets, searching, sorting and aggregation. It also develops the ability to convert raw transactional information into meaningful summaries. From a business perspective, the dashboard can help users identify top products, compare categories, monitor overall sales and understand basic sales performance. The project therefore connects programming concepts with a familiar e-commerce use case and improves both technical and analytical skills.

9. Area of Future Improvement

The project can be expanded by connecting the dashboard to a live database or e-commerce API so that sales information is updated automatically. Advanced filters can be added for date, location, product category and customer segment. Future versions can include real-time charts, customer segmentation, recommendation features and demand forecasting. For very large datasets, indexing, optimized searching and more efficient data structures can reduce processing time. Predictive analytics could also be introduced to forecast product demand and support inventory planning.

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